Adversary Emulation Practice Report

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Title

Adversary Emulation and Detection using MITRE Caldera and Wazuh

Objective

The objective of this task is to simulate adversary techniques using MITRE Caldera and evaluate the effectiveness of SOC detection capabilities using Wazuh. The focus is on emulating a spearphishing attack (MITRE ATT&CK T1566) and validating whether the system can detect and respond to such threats.

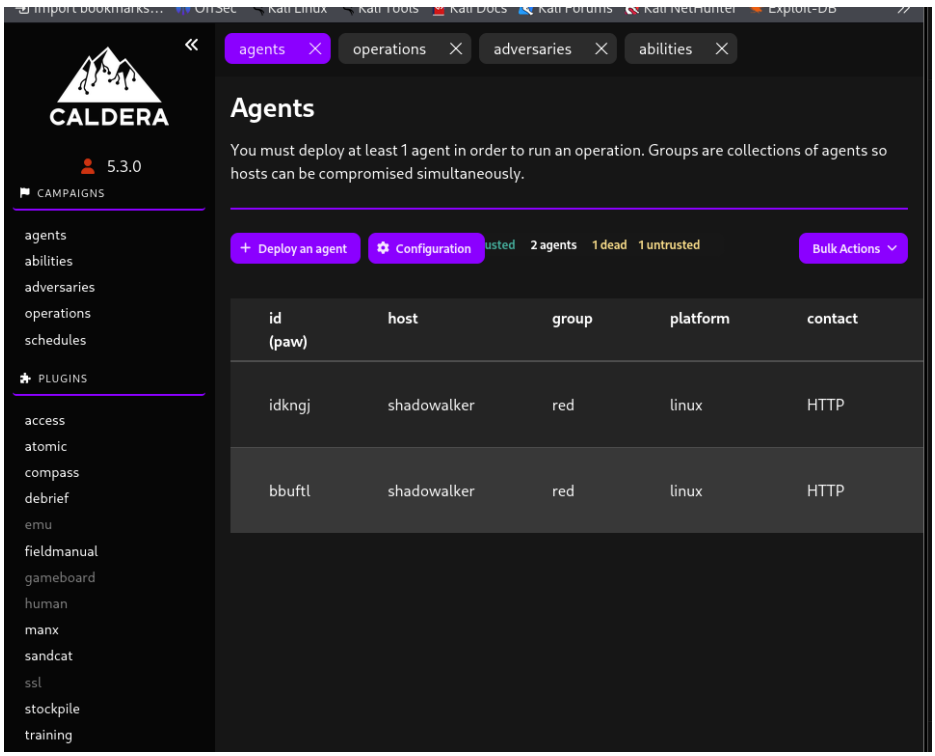
Tools Used

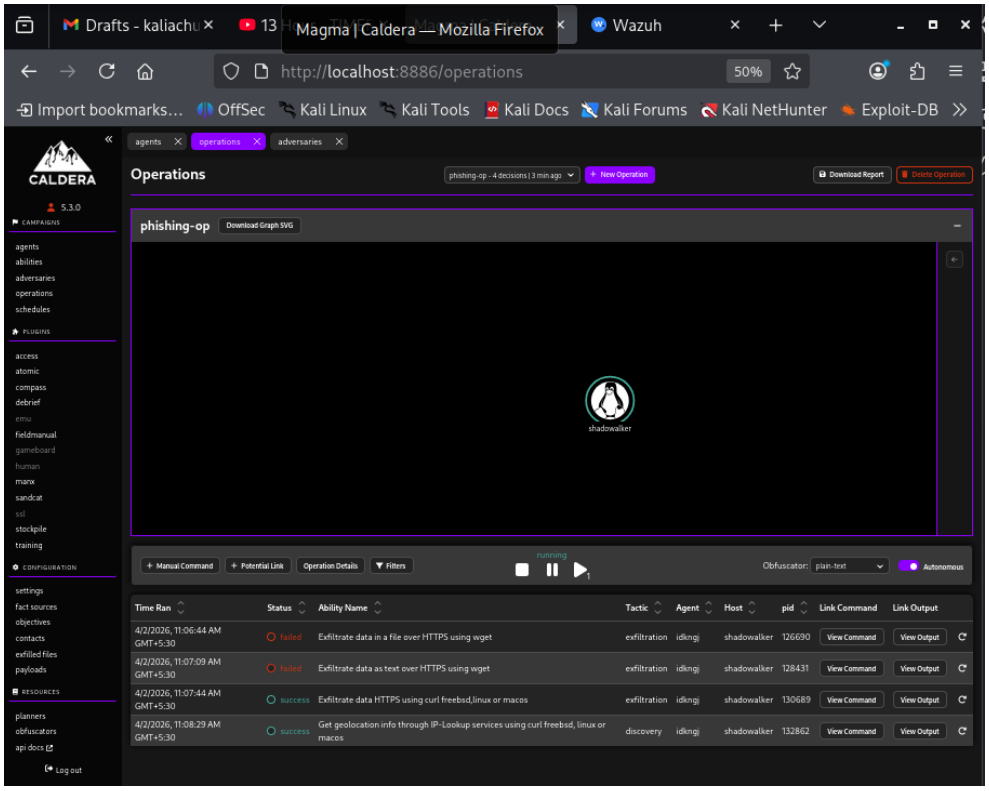
- MITRE Caldera
- Wazuh

Emulation Simulation

A spearphishing attack scenario was simulated using MITRE Caldera in a controlled lab environment. The simulation mimicked a user interacting with a malicious link, which triggered a PowerShell-based download activity on a Windows endpoint. This behavior represents MITRE ATT&CK technique **T1566 (Phishing)**.

Wazuh was configured to monitor Windows event logs and PowerShell activity. Custom detection rules were implemented to identify suspicious commands such as file downloads initiated through PowerShell.



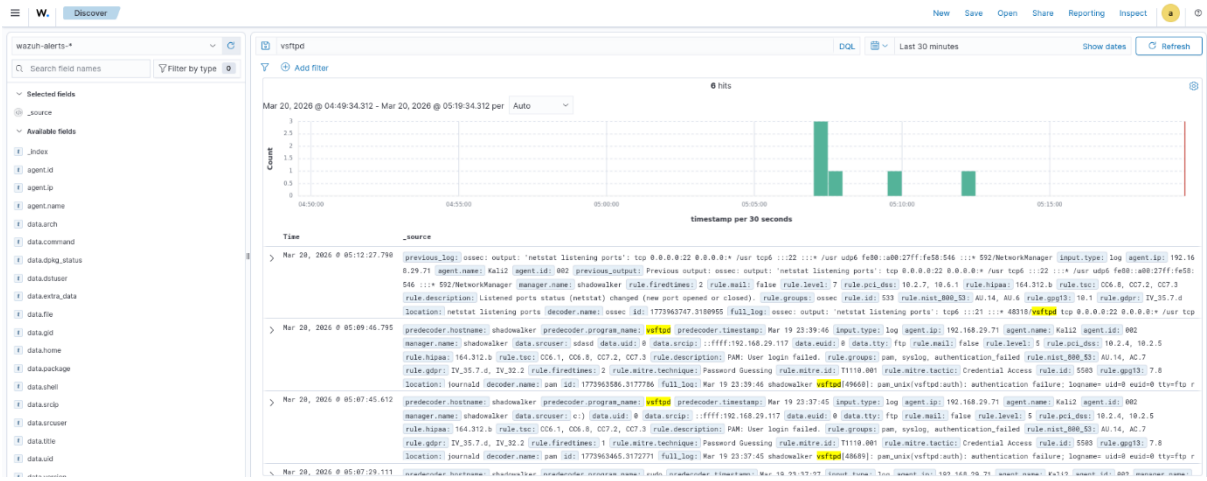


Analysis

- The emulated attack successfully generated suspicious activity on the endpoint.
- Wazuh detected the activity through log monitoring and rule matching.
- Alerts were triggered based on PowerShell download behavior.
- The SOC demonstrated the ability to identify phishing-related techniques.

Emulation Report

The adversary emulation was conducted using MITRE Caldera to simulate a spearphishing attack mapped to MITRE ATT&CK technique T1566. The attack involved a user executing a malicious link, which triggered a PowerShell-based file download on the target system. Wazuh successfully detected the suspicious activity by monitoring system and PowerShell logs and applying custom detection rules. The alert confirmed that the SOC can identify phishing-related threats effectively. However, a detection gap identified was the reliance on custom rules and enhanced logging, as default configurations may not detect such behavior. Improving rule coverage and enabling advanced logging will enhance detection accuracy and response efficiency.



Results

- Attack Simulation: Successful
- Detection: Successful
- TTP Covered: T1566 (Phishing)
- Detection Method: Log monitoring + custom rules

Detection Gaps Identified

- Dependence on custom detection rules
- Limited visibility without PowerShell logging
- No automated response mechanism
- Potential delay in alert correlation

Conclusion

This practical demonstrates the importance of adversary emulation in testing SOC capabilities. MITRE Caldera effectively simulated real-world attack behavior, while Wazuh successfully detected the activity. The exercise highlights the need for proper logging, detection rule tuning, and automation to improve SOC performance.